BLG 478E - NETWORK SECURITY

HOMEWORK

SPRING 2026

1. Objective

The goal of this assignment is to implement a secure communication between Alice and Bob using a Key Distribution Center (KDC). You will utilize some cryptographic primitives for and perform network traffic analysis to validate the integrity and confidentiality of the system.

2. Requirements

- Any programming language for socket programming.
- Wireshark for network analysis.

3. System Architecture

The system consists of three entities:

- KDC (Key Distribution Center): A trusted third party that shares a unique master key with every user.
- Alice (Sender): Initiates the communication request and transmits the secure message.
- Bob (Receiver): Receives the transmission and performs cryptographic verification.

Some notations and meanings specified in Table 1:

Table 1: Notations

Parameter	Meaning
ID_A	Unique identifier for Alice.
$K_{B\text{-}KDC}$	Master key of Bob. Shared between Bob and KDC.
K_S	Session key for secure communication between Alice and Bob.
M	Plaintext message.
$E(K_s, M)$	Ciphertext. M encrypted with K_s.
H(M)	Hash of M.
M || H(M)	M combined with H(M).

- You are free to choose your favorite programming language.
- Each entity will be a different process in your system.
- For symmetric encryption, use AES-256.
- For hashing, use SHA-256.
- You may use libraries for encryption and hash algorithms.
- You need to assign a different port for each entity.
- M must contain your student number. (e.g., M="11111111 Network Sec.")

4. **System Workflow**
   Implement the system given in Figure 1.

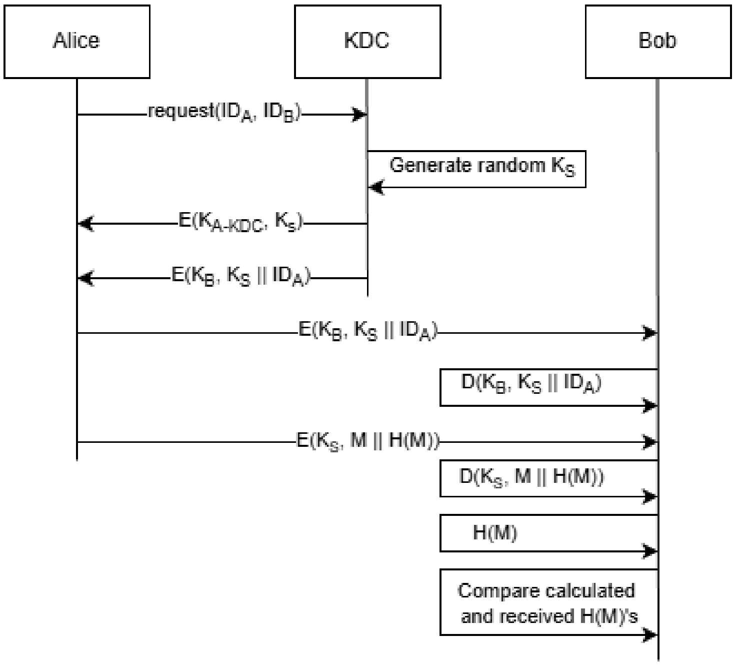

Figure 1: System Workflow

5. **Submission Requirements**
   Source Codes: At least three different source code for KDC, Alice, and Bob. (You may include additional codes depending on your design.)
   Analysis Report: Your report should contain and explain:
   - Baseline Transmission: Screenshot of Wireshark packet capture without any encryption (plaintext message highlighted).
   - Secured Transmission: Screenshot of Wireshark packet capture sent from Alice to Bob (ciphertext message highlighted).
   - Proof of Implementation: Screenshots of each entities terminal step by step. Show the sent and received messages as well as inputs and outputs of encryption and decryption processes. Explicitly define all the keys, plaintexts, hashes, encrypted and decrypted values in the report.
   - Integrity Test: Manually flip 1 bit in the H(M) before transmission and show a screenshot of Bob's terminal throwing a "Verification Failed" error.

   Video Demo (≤ 5 minutes): Your video should cover all the steps in the report.

6. **Grading Rubric**
   - Correct packet transmission between entities: 1 pt
   - Correct implementation of the given workflow: 2 pts
   - Clear report and explanation: 2 pts
   - Video Demo: 2 pts
   - Live Demo: 3 pts (Brief explanation of the work in one-on-one Q&A session.)